

Let V be a fin. dim vector space, and $T: V \rightarrow V$ be a linear map.
Define $\chi_T(t) = \det(tI - [T]_{\mathcal{B}})$ be the characteristic polynomial,
 $m_T(t)$ be the minimal polynomial.

Lemma. For $c \in F$,

$$\chi_T(c) = 0 \text{ if and only if } \ker(T - cI) \neq \{0\}.$$

Proof. $\chi_T(c) = 0 \Leftrightarrow \det(cI - [T]_{\mathcal{B}}) = 0 \Leftrightarrow cI - [T]_{\mathcal{B}}$ is non-singular.

Lemma. For $c \in F$,

$$\chi_T(c) = 0 \text{ if and only if } m_T(c) = 0.$$

Proof. Suppose that $\chi_T(c) = 0$. Then, $T(\alpha) = c \cdot \alpha$ for some non-zero $\alpha \in V$.
Therefore, $0 = m_T(T)(\alpha) = m_T(c) \cdot \alpha$, but since $\alpha \neq 0$, $m_T(c) = 0$.

Conversely, suppose that $m_T(c) = 0$. Then,

$$m_T(t) = (t - c) \cdot g(t) \text{ for some } g(t) \in F[t].$$

By the minimality of m_T , $g(T) \neq 0$, so that there exists $\alpha \in V$ such that $g(T)(\alpha) \neq 0$, thus

$$(T - cI)(g(T)(\alpha)) = m_T(T)(\alpha) = 0$$

So proved. $\square$

The Primary decomposition theorem.

Let T be a linear operator on the finite-dimensional vector space V over a field F .
Let P be the minimal polynomial for T with the irreducible decomposition

$$P = p_1^{r_1} \cdot p_2^{r_2} \cdots p_k^{r_k} = \prod_{i=1}^k p_i(x)^{r_i}.$$

Then,
 $V = \bigoplus_{i=1}^k \ker p_i^{r_i}(T)$ and each $\ker p_i^{r_i}(T)$ is invariant T .

Moreover, the restriction of T by $\ker p_i^{r_i}(T)$ has the minimal polynomial as $p_i^{r_i}$.

Thm. Let α be any non-zero vector in V and let p_α be the T -annihilator of α .

- 1) $\deg p_\alpha = \dim Z(\alpha, T)$
- 2) $Z(\alpha, T) = \text{span} \{T^k \alpha \mid 0 \leq k \leq \deg p_\alpha\}$
- 3) The minimal polynomial of $T|_{Z(\alpha, T)}$ is p_α .

Cyclic decomposition Theorem.

Thm. Let $T: V \rightarrow V$ be a linear operator on finite-dimensional space V .

Suppose that the minimal polynomial of T has a irreducible decomposition

$$m_T(x) = p_1(x)^{r_1} \cdots p_k(x)^{r_k}$$

Then, by the Primary decomposition, V is decomposed by T -invariant spaces

$$V = \ker p_1^{r_1} \oplus \cdots \oplus \ker p_k^{r_k}$$

With $T|_{\ker p_i^{r_i}}$ has minimal polynomial as $p_i^{r_i}$.

Now, for fixed $1 \leq i \leq k$, denote $T_i = T|_{\ker p_i^{r_i}}$ and $W_i = \ker p_i^{r_i}$.

Then, there are non-zero vectors $\alpha_1, \dots, \alpha_n$ such that

$$W_i = Z(\alpha_1, T_i) \oplus \cdots \oplus Z(\alpha_n, T_i)$$

satisfying for some unique integers $1 \leq r_n \leq \cdots \leq r_2 \leq r_1 = r_i$,

$$m_{T_i|_{Z(\alpha_s, T_i)}} = p_i^{r_s}.$$

Primary decomposition: splits case.

Suppose that the minimal polynomial of T splits as

$$m_T(t) = (t - \lambda_1)^{r_1} \cdots (t - \lambda_k)^{r_k}.$$

Then, by the Primary decomposition,

$$V = \ker(T - \lambda_1)^{r_1} \oplus \cdots \oplus \ker(T - \lambda_k)^{r_k}.$$

Now, we get the resolution of identity

$$T = T \circ E_1 + \cdots + T \circ E_k \quad \text{where } E_i \text{ is projection onto } \ker(T - \lambda_i)^{r_i} \\ \text{along } \bigoplus_{j \neq i} \ker(T - \lambda_j)^{r_j}.$$

and let

$$D = \lambda_1 E_1 + \cdots + \lambda_k E_k.$$

Then,

$$N = T - D = (T - \lambda_1 I) \circ E_1 + \cdots + (T - \lambda_k I) \circ E_k.$$

is nilpotent, with the order is at most $\max\{r_1, \dots, r_k\}$.

Moreover, for $i \neq j$,

$$(T - \lambda_i I) \Big|_{\ker(T - \lambda_j)^{r_j}} = \underbrace{(\lambda_j - \lambda_i) \cdot I}_{\text{invertible}} + \underbrace{(T - \lambda_j I) \Big|_{\ker(T - \lambda_j)^{r_j}}}_{\text{nilpotent}}$$

So it is invertible.

Jordan form.

Let $T: V \rightarrow V$ be a linear operator on n -dimensional vector space over F , with $\chi_T(t) = (t - \lambda_1)^{r_1} \cdots (t - \lambda_k)^{r_k}$

and $m_T(t) = (t - \lambda_1)^{s_1} \cdots (t - \lambda_k)^{s_k}$ ($1 \leq s_i \leq r_i, \forall i$.)

1). $\bigcup_{N=1}^{\infty} \ker(T - \lambda_i)^N = \ker(T - \lambda_i)^{s_i}$.

Proof. The $\supseteq$ inclusion is clear. And, by the Primary decomposition, given $x \in \bigcup_{N=1}^{\infty} \ker(T - \lambda_i)^N$, there is unique representation

$$x = x_1 + \cdots + x_k \quad \text{where } x_s \in \ker(T - \lambda_s)^{s_s}, \forall s.$$

Fix $m \in \mathbb{N}$ such that $x \in \ker(T - \lambda_i)^m$. That is,

$$(T - \lambda_i I)^m(x) = \sum_{s=1}^k (T - \lambda_i I)^m(x_s) = 0.$$

Meanwhile, each W_s is $T - \lambda_i I$ invariant and $(T - \lambda_i I)|_{\ker(T - \lambda_s I)^{s_s}}$ is invertible, therefore $x_s = 0$ for all $s \neq i$. Finally, $x = x_i \in \ker(T - \lambda_i)^{s_i}$. So proved. $\square$

$\exists x_1, x_2, \dots, x_{s_i}$ not independent.

2). $\dim \ker(T - \lambda_i)^{s_i} = \dim \bigcup_{N=1}^{\infty} \ker(T - \lambda_i)^N = r_i$.

Proof. By the Primary decomposition, $V = \bigoplus \ker(T - \lambda_i)^{s_i}$, so that

$$\chi_T(t) = \prod_{i=1}^k \chi_{T|_{\ker(T - \lambda_i)^{s_i}}}(t) = \prod_{i=1}^k (t - \lambda_i)^{\dim \ker(T - \lambda_i)^{s_i}}, \quad \text{So proved. } \square$$

In $\ker(T - \lambda_i)^{s_i}$, it has only λ_i as eigenvalue.

3). $s_i =$ largest Jordan block size for λ_i .

$\dim \ker(T - \lambda_i I) =$ number of λ_i -blocks.

Proof. $J_{n_1}(\lambda) \oplus \cdots \oplus J_{n_g}(\lambda)$, then $m_{T|_V}(t) = \text{lcm}((t - \lambda)^{n_1}, \dots, (t - \lambda)^{n_g}) = (t - \lambda)^{\max n_i}$.

Summary.

Jordan form.

Let $T: V \rightarrow V$ be a linear operator on n -dimensional vector space over F ,

with $\chi_T(t) = (t - \lambda_1)^{r_1} \cdots (t - \lambda_k)^{r_k}$

and $m_T(t) = (t - \lambda_1)^{s_1} \cdots (t - \lambda_k)^{s_k} \quad (1 \leq s_i \leq r_i, \forall i.)$

Then, the Jordan form is: $s_{i1} \geq s_{i2} \geq s_{i3} \geq \dots$,

$$J = \begin{bmatrix} J_{s_1}(\lambda_1) & & & \\ & J_{s_{i2}}(\lambda_1) & & \\ & & \ddots & \\ & & & J_{s_{i m_i}}(\lambda_1) & \\ & & & & J_{s_2}(\lambda_2) & \ddots \end{bmatrix}$$

r_1

Computations.

$\mathbb{H}K$, $\eta, 3, 3$,

Let $A \in M_{5 \times 5}(\mathbb{C})$ with

$$\chi_A(t) = (t-2)^3 \cdot (t+\eta)^2 \quad \text{and} \quad m_A(t) = (t-2)^2 \cdot (t+\eta).$$

Determine the Jordan form of A .

Proof. Since $(t+\eta)$ has multiplicity 2 in χ_A and 1 in m_A , the Jordan form of A has $[-\eta] \oplus [-\eta]$ as a block corresponding to $-\eta$, and $(t-2)$ has multiplicity 2 in m_A , it has a Jordan block

$$J_2(2) = \begin{bmatrix} 2 & 1 \\ & 2 \end{bmatrix}$$

and $(t-2)$ has multiplicity 1 in χ_A , the last block is forced $[2]$. Consequently, the Jordan form is

$$J_A = \begin{bmatrix} 2 & 1 & & & \\ & 2 & & & \\ & & 2 & & \\ & & & -\eta & \\ & & & & -\eta \end{bmatrix}$$

TKAIST, 2027 spring.

Let $A \in M_{8 \times 8}(\mathbb{R})$. Suppose $\chi_A(x) = (x+1)^5 \cdot (x-2)^3$

1). Assume

$\dim \ker(A+I) = 2$, $\dim \ker(A+I)^2 = 4$, $\dim \ker(A+I)^3 = 5$
 $\dim \ker(A-2I) = 1$, $\dim \ker(A-2I)^2 = 2$, $\dim \ker(A-2I)^3 = 3$.
Determine all possible Jordan Canonical form of A .

2). Assume

$$m_A(x) = (x+1)^3 \cdot (x-2)^3.$$

Determine all possible Jordan Canonical form of A .

Proof. 1). Since $\dim \ker(A+I)^3 = 5$ equals to the multiplicity of $x+1$ in χ_A ,

$$\mathbb{R}^8 = \ker(A+I)^3 \oplus \ker(A-2I)^3. \text{ and, moreover,}$$

$\dim \ker(A+I) = 2$ implies A has two Jordan blocks for -1 ,

$\dim \ker(A-2I) = 1$ implies A has only one Jordan block for 2 .

Moreover, $\dim \ker(A+I)^2 - \dim \ker(A+I) = 2$ but

$\dim \ker(A+I)^3 - \dim \ker(A+I)^2 = 1$ therefore it has Jordan blocks of size 2 and 3. Finally, there is only one possible form

$$J_A = J_3(-1) \oplus J_2(-1) \oplus J_3(2).$$

2). Since $(x+1)$ has multiplicity 3 in m_A , there are possible partitions

$$5 = 3 + 2 = 3 + 1 + 1,$$

and since $x-2$ has multiplicity 3 in m_A , there are two possible forms:

$$J_A = J_3(-1) \oplus J_2(-1) \oplus J_3(2) \text{ or}$$

$$= J_3(-1) \oplus [-1] \oplus [-1] \oplus J_3(2)$$

$\square$

HK, 7.3, 15 ~ 16.

Let $N \in M_{\text{non}}(\mathbb{C})$ be a nilpotent matrix over Complex numbers.

For any non-zero $c \in \mathbb{C}$, $cI + N$ has a square root.

Moreover, deduce that every invertible matrix over $\mathbb{C}$ has a square root.

Proof. Let $A = c \cdot \left(\sum_{k=0}^{n-1} \frac{1}{2^k} \cdot N^k \right)$.